MicroSens TX-4000: Multi-Axis Inertial Sensor Datasheet

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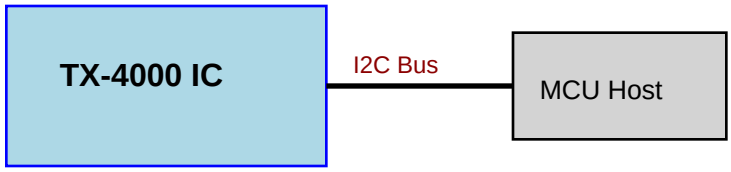
1. Pinout Configuration & Terminal Functions

Pin #	Pin Name	Signal Type	Voltage Level	Description
1	VDD	Power	3.3 V DC	Main positive supply input for internal logic
2	GND	Power	0.0 V	Ground reference for digital and analog circuits
3	SCL	Digital Input	3.3 V I2C	I2C Serial Clock bus line with internal pullup
4	SDA	Digital I/O	3.3 V I2C	I2C Serial Data bus bidirectional line
5	INT1	Digital Output	3.3 V Push-Pull	Interrupt 1 output for threshold alarm events
6	V_SENS	Power Input	5.0 V Analog	Dedicated analog sensor rail for high-precision ADC
7	SYNC	Digital Input	3.3 V	External hardware sample synchronization pin
8	RESET_N	Digital Input	3.3 V Active-Low	Master reset line: pull low to reset core



2. Absolute Maximum Ratings & Operating Conditions

Parameter	Symbol	Min	Typ	Max	Unit
Supply Voltage (VDD)	VDD_MAX	-0.3	3.3	4.0	V
Sensor Rail Voltage	VSENS_MAX	-0.3	5.0	5.5	V
Operating Temperature	TOPER	-40	+25	+105	°C
Storage Temperature	TSTG	-55	--	+150	°C
I2C Clock Frequency	FSCL	10	400	1000	kHz
Standby Current	IDD_STBY	--	2.5	8.0	µA
Active Current Consumption	IDD_ACTIVE	--	1.2	2.8	mA